Requirement Analysis

Data Flow Diagrams and User Stories

Date	02 October 2026
Team ID	SWTID-2026-5170
Project Name	Script-Controlled ACL – Restrict Record Access Based on Custom Script
Maximum Marks	4 Marks

Data flow diagram (Level 0 / Level 1)

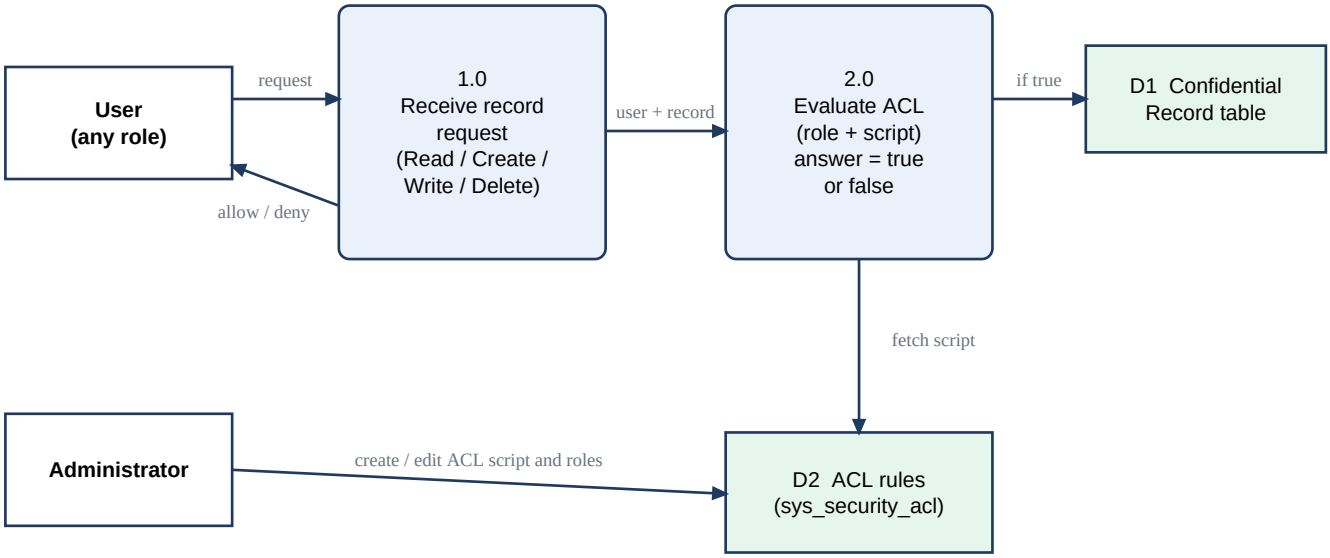

Figure 1: Data flow of a record request through the script-controlled ACL

The user's request is received by the platform, the ACL engine fetches the ACL rules for the table and operation, runs the script, and only if the script returns **true** is the record in D1 returned or changed. Otherwise the request is denied.

User stories

Functional requirement (Epic)	USN	User story / task	Priority	Release
Milestone-1: Users and Roles	USN-1	As an administrator, I can create the project users (admin, owner, standard user, no-role user).	High	Sprint-1
Milestone-1: Users and Roles	USN-2	As an administrator, I can create the roles u_record_admin and u_record_user and assign them to users.	High	Sprint-1
Milestone-2: Tables Creation	USN-3	As a developer, I can create the Confidential Record table with the required fields.	High	Sprint-1
Milestone-2: Tables Creation	USN-4	As a tester, I can add sample records (confidential and non-confidential) owned by different users.	High	Sprint-1
Milestone-3: ACL - READ	USN-5	As a security admin, I can create a script-controlled READ ACL so users see only permitted records.	High	Sprint-1
Milestone-3: ACL - READ	USN-6	As a tester, I can impersonate each user and verify the list and form visibility.	High	Sprint-1

Functional requirement (Epic)	USN	User story / task	Priority	Release
Milestone-4: ACL - CREATE	USN-7	As a security admin, I can create a script-controlled CREATE ACL for authorised roles only.	High	Sprint-2
Milestone-5: ACL - WRITE	USN-8	As a security admin, I can create a script-controlled WRITE ACL that lets only owners/admins edit records.	High	Sprint-2
Milestone-6: ACL - DELETE	USN-9	As a security admin, I can create a DELETE ACL that restricts deletion to administrators.	High	Sprint-2
Milestone-7: Conclusion	USN-10	As a tester, I can run the full access matrix for Read, Create, Write and Delete.	High	Sprint-2
Milestone-7: Conclusion	USN-11	As a project team, we can document the implementation, testing and results.	Medium	Sprint-2
Milestone-7: Conclusion	USN-12	As a project team, we can record the demo video and publish the GitHub repository.	Medium	Sprint-2